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Late Cretaceous Paleomagnetism and Clockwise Rotation of the Silver Bell Mountains, South Central Arizona

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Late Cretaceous ash flow volcanism in the Silver Bell Mountains of southern Arizona (32.3°N, 248.5°E) was associated with caldera formation and porphyry copper mineralization. Oriented samples from 34 sites in volcanic, volcaniclastic, and intrusive units related to this episode of igneous activity (73–69 Ma) yield a mean paleomagnetic direction of $I = 61.2°$, $D = 24.0°$, $\alpha_{95} = 7.6°$. Primary remanent magnetizations are indicated by the presence of both normal and reversed polarities and by the significantly improved grouping (95% confidence level) of site-mean directions for a widespread tuff unit after structural correction. A comparison of this direction with a direction at Silver Bell calculated from contemporaneous units for stable North America ($I = 58.8°$, $D = 342.3°$, $\alpha_{95} = 7.7°$) indicates a Late Cretaceous paleolatitude anomaly of $-2.4° \pm 7.6°$ (not significant at the 95% confidence level) and a declination anomaly of $41.7° \pm 14.3°$ for the Silver Bell Mountains. Previously determined paleomagnetic data for southeastern Arizona suggest that this apparent clockwise rotation in the Silver Bell Mountains is a local phenomenon. Although preliminary, the average paleomagnetic direction for Oligocene and lower Miocene rocks in the Silver Bell area ($I = 43.8°$, $D = 357.3°$, $\alpha_{95} = 16.5°$) is similar to that calculated for stable North America ($I = 50.2°$, $D = 352.2°$, $\alpha_{95} = 3.9°$), implying that the observed rotation in the Silver Bell Mountains occurred before 26 Ma and was most likely associated with dextral strike-slip movement along the Ragged Top and related WNW trending faults bounding the Silver Bell Mountain block. These data, in conjunction with plate reconstructions and other paleomagnetic data from southwestern North America, imply that WNW trending strike-slip faults may have played an important role during Late Cretaceous to early Tertiary deformation in southern Arizona.

INTRODUCTION

Few paleomagnetic data for rocks of Late Cretaceous to early Tertiary (Laramide) age are available from the southern Basin and Range province of western North America (Figure 1). Paleomagnetic data determined for the Upper Cretaceous Roskruge Volcanics and the lower Paleocene Gringo Gulch Volcanics of southern Arizona [*Vugteveen et al.*, 1981] are basically consistent with the North American apparent polar wander (APW) path but are limited and may not be representative of the entire region. More extensive paleomagnetic studies have been undertaken in Jurassic volcanic rocks from southeastern Arizona [*Kluth et al.*, 1982; *May and Butler*, 1986; *May et al.*, 1986; *May and Butler*, 1987] to help constrain the reference path for ancestral North America during this period of rapid apparent polar wander. Of fundamental importance to these studies is the assumption that southern Arizona has remained stationary with respect to the North American craton since Jurassic time. The primary justification of this assumption has been the general consistency of the available paleomagnetic data for southern Arizona with the North American APW path; discordant results [*May and Butler*, 1987] have been attributed to an anomalous geomagnetic field. However, the southern Basin and Range province is a tectonically disturbed area which, since middle Cretaceous time, has been subjected to Late Cretaceous to early Tertiary compression, large-scale extension in the middle Tertiary, and younger phases of Basin and Range extension along steeply dipping normal faults.

Geological and geophysical investigations imply that much of the North American Cordillera is composed of tectonostratigraphic terranes which are allochthonous with respect to the stable craton [*Coney et al.*, 1980]. Paleomagnetic results for many of these terranes are consistently discordant with the North American APW path, indicating that they have undergone either northward translations or clockwise rotations, or both, relative to the North American continent [*Beck*, 1980]. Many of these displacements are probably caused by large-scale deformation associated with dextral shear between the North American plate and oceanic plates to the west since at least Late Cretaceous time [*Beck*, 1986]. In the Pacific Northwest a monotonic westward increase in the amount of rotation of crustal fragments argues in favor of distributed dextral shear as the cause of the rotations [e.g., *Wells and Heller*, 1988]. The proximity of southern Arizona to the western margin of North America implies that this region may also have been subjected to a similar style of deformation. Paleomagnetic data for the Baja California Peninsula to the southwest indicate that it has been moved northward ($\simeq 9°$) and rotated clockwise ($\simeq 24°$), relative to North America, between Cretaceous and Miocene time [e.g., *Hagstrum et al.*, 1985, 1987*a*]. Furthermore, geologic [*Hamilton*, 1981] and paleomagnetic evidence [*Steiner*, 1986; *Bryan and Gordon*, 1986] indicate that the Colorado Plateau to the northeast rotated clockwise by a smaller amount

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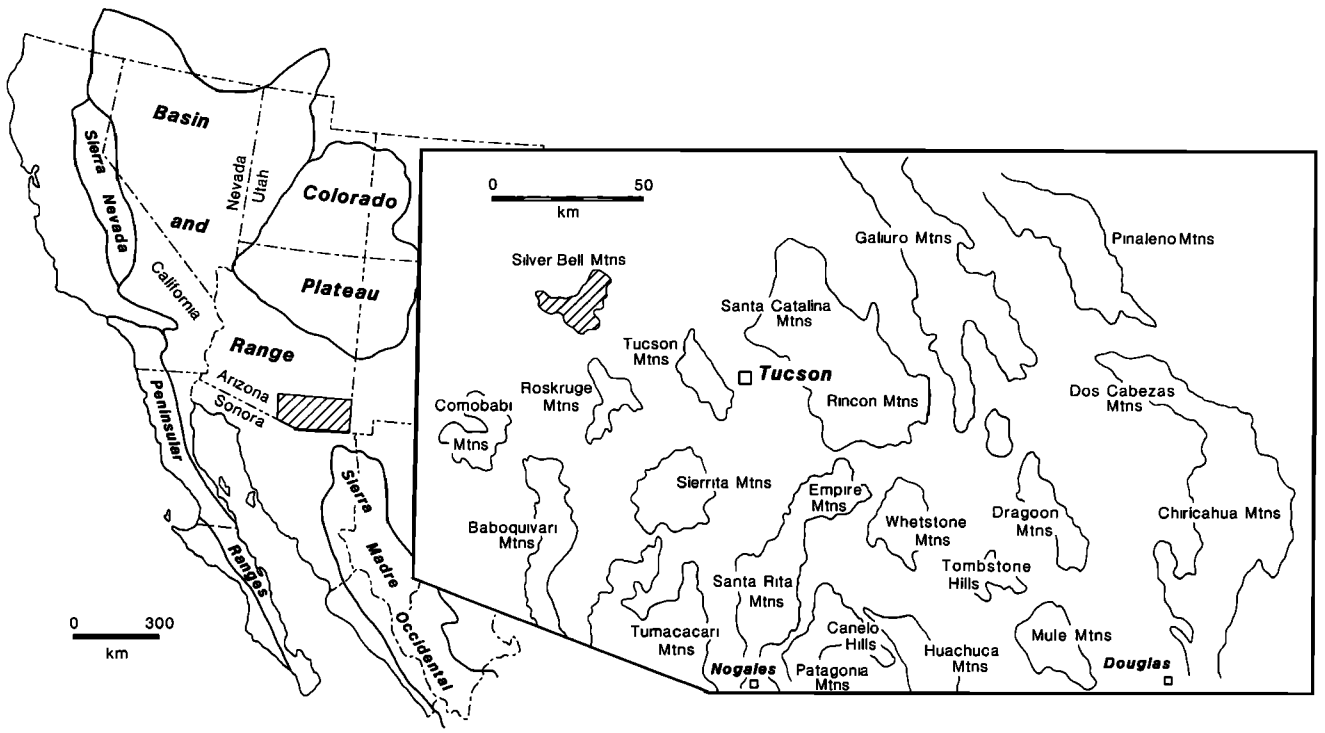


Fig. 1. Regional map on left showing physiographic provinces in southwestern North America [after *Stewart*, 1978] and index map on right showing location of the Silver Bell Mountains in southeastern Arizona.

(4°–11°) after middle Cretaceous time. Thus it appears that vertical-axis rotations with respect to cratonic North America could have played a significant role in the tectonic deformation of the intervening southern Basin and Range province as well.

Paleomagnetic sampling for this study was undertaken in the Silver Bell Mountains of south central Arizona (Figure 1) to obtain additional Late Cretaceous data for the region. Additional paleomagnetic data for southern Arizona would contribute to an improved understanding of the region's tectonic history and its relationship to large-scale plate interactions along the western margin of North America and bear on the question of whether or not the southern Basin and Range province has remained stationary with respect to stable North America since Late Cretaceous time. The discordance of the latest Cretaceous data reported in this paper, with respect to the well-determined North American APW path for this time, most likely indicates that portions of the region have rotated relative to the stable craton. Tertiary lava flows and dikes in the area were also sampled for paleomagnetic analysis in an attempt to constrain the timing of deformation. Preliminary data for these rocks imply that rotation in the Silver Bell Mountains was associated with Late Cretaceous to early Tertiary right-lateral strike-slip faulting rather than with middle Tertiary extension.

Geologic and Tectonic Setting

Late Cretaceous igneous activity and porphyry copper mineralization in the Silver Bell Mountains (Figure 2) were associated with caldera volcanism [*Lipman and Sawyer*, 1985; *Sawyer*, 1987]. The oldest Upper Cretaceous igneous unit in the district is the precaldera biotite granite of El Tiro which has an U-Pb zircon age of 72.7 ± 1 Ma (E. James and D. Sawyer, unpublished data, 1983). The northeast margin of the biotite granite is in complex fault contact with intracaldera tuff of Confidence Peak along the southwestern ring fracture zone of the Silver Bell caldera. Intrusive portions of this contact suggest that the granite may still have been partially molten during caldera subsidence [*Sawyer*, 1987]. Ash flows of the tuff of Confidence Peak were erupted and ponded to a thickness greater than 1.5 km within the Silver Bell caldera, concurrent with its collapse; large blocks of Paleozoic sedimentary rocks enclosed within the intracaldera tuff are interpreted as caldera-collapse megabreccia [*Sawyer*, 1987]. A 250-m-thick section of five small-volume pyroclastic flows and air fall deposits with interbedded volcaniclastic sediments in the western Silver Bell Mountains is correlated with the intracaldera tuff of Confidence Peak on the basis of stratigraphic position and petrochemical affinities [*Sawyer*, 1987].

Depositionally overlying the tuff of Confidence Peak are the moat-filling volcaniclastic sediments of the Claflin Ranch Formation. These sediments are stratigraphically beneath or locally interfinger with the Silver Bell Formation. The Silver Bell Formation includes a diverse sequence of andesitic to dacitic lava flows, domes, hypabyssal intrusions, and related volcaniclastic debris flows and sediments. Granodiorite porphyry plutons intruding the Claflin Ranch Formation and the tuff of Confidence Peak probably represent the cores of source volcanoes for the volcanics of the Silver Bell Formation [*Sawyer*, 1987]. Petrologic, age, and paleomagnetic data [*Mauger et al.*, 1965; D. Sawyer et al., unpublished data, 1988] indicate that the tuff unit of the Mount Lord Volcanics [*Watson*, 1964] in the Silver Bell Mountains [*Sawyer*, 1987] and the Cat Mountain Rhyolite of the Tucson Mountains (Figure 1) are correlative. The Cat Mountain Rhyolite is the

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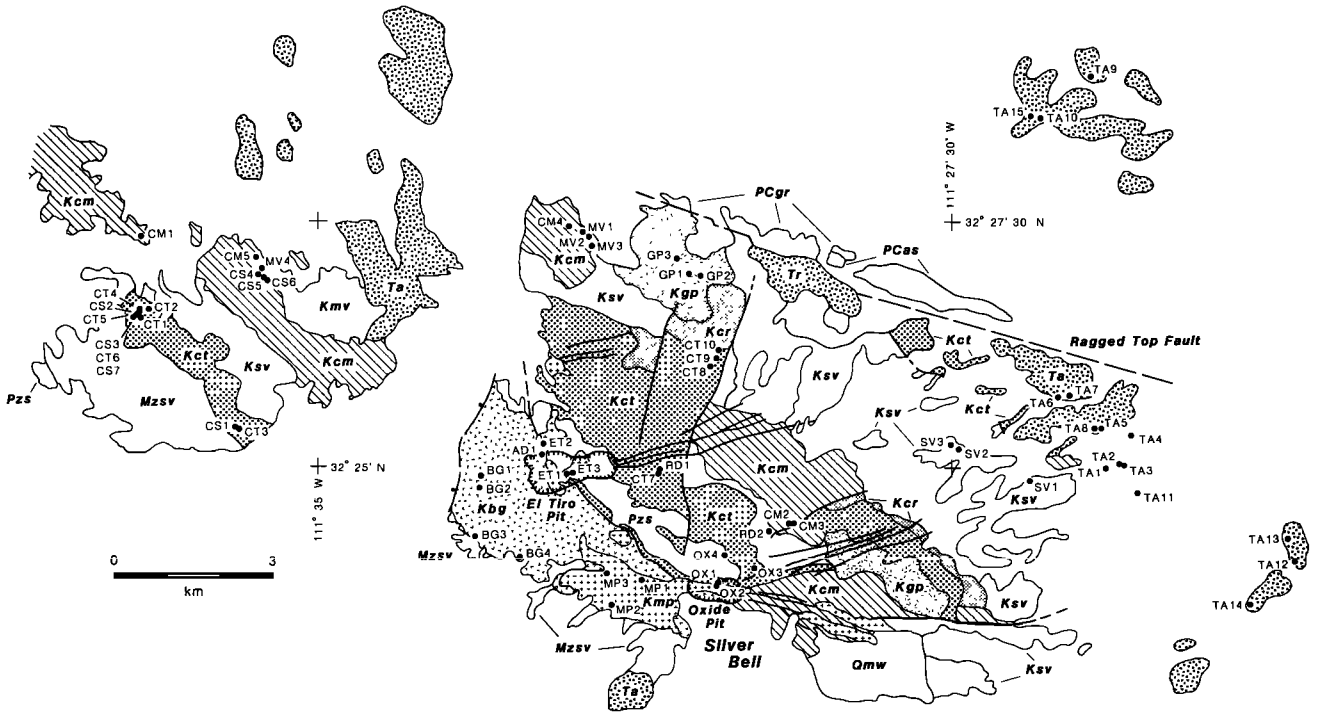


Fig. 2. Geologic map of the Silver Bell and West Silver Bell Mountains [after *Sawyer*, 1987] showing paleomagnetic sampling sites (see Table 1). Map Symbols are PCgr, Precambrian 1.4 Ga granite; PCas, Precambrian Apache Group sedimentary rocks; Pzs, Paleozoic sedimentary rocks; Mzsv, Mesozoic sedimentary and volcanic rocks; Kct, Cretaceous tuff of Confidence Peak and vocaniclastic sedimentary rocks; Kcr, Cretaceous Claflin Ranch Formation; Ksv, Cretaceous Silver Bell Formation; Kgp, Cretaceous granodiorite porphyry; Kmv, Cretaceous Mount Lord Volcanics; Kcm, Cat Mountain Rhyolite; Kmp, Cretaceous quartz monzodiorite porphyry suite and monzodiorite porphyry; Ta, Tertiary andesite and basaltic andesite; Tr, Tertiary rhyolite of Ragged Top; Qmw, Quaternary mine waste. Lines trending NNE from the Oxide and El Tiro pits and to the north are dikes of Cretaceous age.

youngest volcanic unit in the Silver Bell district with a K-Ar sanidine age of 60.9 ± 1.8 Ma [*Mauger et al.*, 1965; *Reynolds et al.*, 1986], and it originated from a caldera in the Tucson Mountains and ponded within the Silver Bell depression. The final Cretaceous magmatism in the Silver Bell district was intrusion of rocks associated with the major porphyry copper deposits. The initial phase of this episode was emplacement of a monzodiorite porphyry pluton and related dikes mostly in the vicinity of the Oxide mineralized center. The intrusion of a suite of plutons ranging in composition from quartz monzodiorite porphyry through granodiorite, consisting mostly of the former, followed along the southwest ring fracture zone of the Silver Bell caldera. K-Ar biotite dates for the quartz monzodiorite porphyry suite range from 68.7 ± 2.0 to 64.8 ± 2.2 Ma [*Mauger et al.*, 1965; *Reynolds et al.*, 1986]. Thus the Laramide igneous activity in the Silver Bell Mountains is bracketed between approximately 73 and 69 Ma.

During Tertiary time, dikes were emplaced in the main Silver Bell range, and basaltic andesite to rhyolitic extrusive flows and domes were deposited on the north and east sides of the range. Basalt, basaltic andesite, and hornblende andesite flows to the east and northeast (Figure 2) are remnants of a small middle Tertiary volcanic field [*Eastwood*, 1970]. Samples from exposures of these rocks just east of the Oxide pit yield a K-Ar biotite age of 28.5 ± 1.4 Ma [*Mauger et al.*, 1965; *Reynolds et al.*, 1986]. The Ragged Top peak north of the Silver Bell range is a flow-banded rhyolite dome inferred from a K-Ar biotite age to have been emplaced along the Ragged Top fault at about 25.6 ± 1.0 Ma [*Mauger et al.*, 1965; *Reynolds et al.*, 1986]. NNW trending rhyolite dikes which cut the main part of the Silver Bell range are thought to be equivalent with the Ragged Top dome based on their lithologic similarity [*Joseph*, 1982]. Biotite andesite dikes were also emplaced along NW trending fractures exposed within the El Tiro pit and have a K-Ar biotite age of 25.3 ± 0.6 Ma (M. Shafiqullah, written communication, 1985).

The N75°–80°W near-vertical Ragged Top fault (Figure 2) is a major crustal discontinuity and juxtaposes terranes with different geologic histories along its length. Although significant dip-slip motion is required to bring Precambrian rocks north of the fault in contact with Cretaceous volcanic and plutonic rocks to the south, major strike-slip displacement is also required. Steps on slickensides within the fault zone, a series of small normal faults to the south oriented N15°–20°E, and north stepping offsets westward along the fault are all consistent with a right-lateral sense of shear [*Sawyer*, 1987]. The Ragged Top rhyolite dome was emplaced across the largest north stepping offset of the fault but is not cut by the fault. To the west of Ragged Top peak the Ragged Top fault truncates an Upper Cretaceous granodiorite porphyry pluton. Thus the age of fault movement is constrained between Late Cretaceous (69–73 Ma) and middle Tertiary (26 Ma) time. WNW trending faults, probably strike-slip, also bound the Silver Bell Mountains to the south and separate them from the Waterman Mountains, a small range

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situated between the Silver Bell and the Roskruge Mountains.

Northwest trending subvertical faults with probable strike-slip displacement are a major element of late Mesozoic tectonics throughout southeastern Arizona [*Titley*, 1976; *Drewes*, 1981]. Displacements associated with these faults are difficult to interpret because movement along them has been recurrent and diverse. *Swan* [1976] mapped a WNW trending fault zone in the Stockton Pass area of the Pinaleno Mountains which apparently had at least 6 km of left-lateral movement during Precambrian time. Faults within the same zone apparently underwent right-slip movement in post-Oligocene time as well [*Thorman*, 1981]. *Drewes* [1972, 1981] infers early Tertiary left-lateral offset for the Sawmill Canyon fault in the Santa Rita Mountains and for the Apache Pass fault in the northern Chiricahua Mountains. In southwestern New Mexico, *Seager* [1983] identified Laramide right-lateral strike-slip faulting along WNW and NW trending faults related to basement-cored uplifts. Based on regional dike and fracture patterns, a tectonic stress field inferred for southern Arizona during Laramide time [*Rehrig and Heidrick*, 1976; *Heidrick and Titley*, 1982] suggests that the NW trending strike-slip faults would have been subjected to left-lateral shear. *Keith and Wilt* [1986] confine the left-lateral movement on these faults to a medial phase (65–55 Ma) of the Laramide orogeny which is also characterized by epizonal plutonism and porphyry copper mineralization.

The Laramide orogeny in southeastern Arizona also included an earlier phase of NE–SW oriented compression, but the style and distribution of the compressional event is controversial as well. *Drewes* [1978, 1981] inferred that the style of thrust faulting was continuous with the regional thin-skinned transport of the northern and central Cordillera and of the Sierra Madre Oriental to the south in Mexico. In contrast, *Davis* [1979] concluded that the thrust and reverse faults of southern Arizona were more localized, bounded basement-cored uplifts, and formed along older NW trending flaws in the pre-Late Cretaceous rocks. The breadth of the Laramide orogenic zone in western North America has been attributed to low-angle subduction and coupling of the Farallon plate with the overlying continental lithosphere [*Lipman et al.*, 1971; *Dickinson and Snyder*, 1978]. Reconstructions of plate motions for Laramide time [*Engerbretson et al.*, 1984] indicate a period of rapid convergence between the Farallon and North American plates. These reconstructions also allow significant northward motion of the Farallon plate relative to North America and show that the Kula–North America motion had an even larger component of northward displacement. It is uncertain, however, which plate, the Farallon or Kula, interacted with the southwestern margin of North America in Late Cretaceous to early Tertiary time [*Page and Engebretson*, 1984].

The southern Basin and Range province underwent two distinct episodes of extensional deformation in middle to late Tertiary time. The older episode occurred between about 25 and 13 Ma [*Coney*, 1980] and is characterized by extreme extension associated with regional low-angle normal or detachment faults. These low-angle faults, where exposed, commonly separate brittlely deformed upper plate supracrustal rocks from ductiley deformed lower plate mylonitic gneisses [*Davis*, 1980, 1983]. Although detachment faults have not been observed in the Silver Bell Mountains, the uniform tilt (30°–35° ENE) of the range and, similarly, of the Tucson Mountains to the southeast (Figure 1) suggests that they may be situated above a regional detachment fault which dips westward from the Rincon–Santa Catalina–Tortolita and Picacho core complexes to the east [*Davis*, 1980; *Rehrig and Reynolds*, 1980; *Coney and Reynolds*, 1980]. The younger Pliocene to Pleistocene Basin and Range extension, characterized by high-angle normal faults, is largely responsible for the present bedrock topography of deep alluvial basins separating northward trending mountain ranges [*Eberly and Stanley*, 1978].

METHODS

Sampling for this study was undertaken in the Silver Bell Mountains because the thick sequence of Upper Cretaceous rock units exposed there is suitable for determining a reliable paleomagnetic pole for southern Arizona. The sequence of rocks spans enough time to average secular variation of the geomagnetic field and is exposed over a large enough area that proximal effects of hydrothermal alteration related to copper mineralization can be avoided. In addition, errors associated with local tectonism are most likely minimized due to the relatively simple structure of the mountain range. Generally six to eight samples were collected at each of 45 sites in the Upper Cretaceous volcanic, volcaniclastic, and intrusive units (Table 1 and Figure 2). Samples from an additional 17 sites were collected in middle Tertiary lava flows and dikes exposed to the east and within the main Silver Bell range (Table 2 and Figure 2). The samples were taken as 2.5-cm drill cores and were oriented in the field, with an estimated accuracy of 2°, using magnetic and solar compasses and a clinometer. Attitudes for the extrusive rock units were based on interbedded sedimentary rocks or eutaxitic structures in pyroclastic rocks. An estimate of attitude for the intrusive units was made by averaging 45 measurements on overlying layered units from throughout the eastern Silver Bell Mountains. Structural corrections were made by a simple rotation of the layering plane to the horizontal about the present strike axis.

Cylindrical specimens 2.2 cm in length were cut from the core samples and were subjected to either progressive alternating field (af) or thermal demagnetization. Detailed af demagnetizations were done on one or two pilot specimens from each site and served to determine the demagnetization steps for the site's remaining specimens. The af demagnetization in peak fields between 1 and 100 mT was usually successful in isolating the characteristic component of magnetization for the samples. Specimens from sites where pilot af treatments proved ineffective were subjected to stepwise thermal demagnetization in air to maximum temperatures of 680°C. The remanent magnetization of individual specimens was measured with a cryogenic magnetometer.

Directions of the characteristic magnetizations for each sample were determined from demagnetization data using a least squares method based on principal component analysis [*Kirschvink*, 1980]. For samples with single-component magnetizations, least squares lines were fitted to the demagnetization data. Least squares planes were fitted to the data for samples having two overlapping components of magnetization. The characteristic magnetizations for sites with overlapping components were determined from the intersections of demagnetization planes for the individual samples [*Halls*,

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TABLE 1. Site-Mean Directions and Poles for Upper Cretaceous Units From the Silver Bell Mountains, Arizona

Site	Uncorrected		Strike/Dip, deg	Corrected		N	R	k	α_{95}, deg	Pole	
	I, deg	D, deg		I, deg	D, deg					Latitude, deg	Longitude, deg
Quartz Monzodiorite Porphyry											
MP1	60.7	349.8	330/29	43.1	21.0	6	5.9846	324	3.7	70.3	355.0
MP2	75.8	53.7	330/29	46.9	57.8	6	5.8307	30	12.5	40.5	327.4
MP3	58.8	345.1	330/29	43.1	16.9	8	7.9508	142	4.7	73.5	0.7
Quartz Monzodiorite Porphyry: Oxide Pit											
OX1	40.6	352.6	330/29	25.4	9.1	7	6.6223	16	15.6	69.2	42.7
OX2	48.6	348.8	330/29	33.6	11.3	8	7.7426	27	10.8	72.7	30.0
OX3	53.1	10.6	330/29	30.7	28.0	10	9.7451	35	8.2	60.1	3.9
OX4	44.9	355.2	330/29	28.2	13.4	6	5.9384	81	7.5	68.8	30.4
Quartz Monzodiorite Porphyry: El Tiro Pit											
ET1	80.9	278.7	330/29	67.5	45.0	8	7.9950	1389	1.5	52.5	296.5
ET2	76.7	140.2	330/29	56.3	84.1	8	7.9678	217	3.8	23.0	308.3
ET3	73.1	299.1	330/29	65.4	23.3	8	7.9932	1026	1.7	66.8	291.1
Cat Mountain Rhyolite											
CM1	−68.7	145.1	291/36	−39.8	178.0	6	5.9949	988	2.1	80.0	79.3
CM2	−48.8	157.2	328/30	−36.8	183.7	6	5.9781	229	4.4	77.7	52.1
CM3	−37.8	163.8	350/35	−33.4	189.7	6	5.9932	733	2.5	73.4	34.5
CM4	−79.3	232.9	278/38	−43.9	198.5	6	5.9771	218	4.6	72.5	356.4
CM5	−79.9	112.3	268/45	−40.2	166.0	7		[−55,−1]	[6.5, 8.3]	74.4	124.3
Average				−39.3	183.2	5		[−110,−24]	[4.3, 9.1]	79.5	52.3
Mount Lord Volcanics											
MV1	−62.4	251.7	253/37	−44.5	203.5	8	7.9488	137	4.8	68.6	349.5
MV2	−88.9	67.6	253/40	−50.1	161.3	7	6.9816	326	3.4	74.0	157.8
MV3	−81.2	233.8	248/40	−47.1	170.5	4	3.9960	754	3.4	80.9	134.1
MV4	−57.1	198.1	286/42	−15.1	197.2	8	7.9626	187	4.1	60.5	31.8
Granodiorite Porphyry											
GP1	71.5	5.1	330/29	47.8	37.2	8	7.9072	75	6.4	57.9	334.9
GP2	78.9	15.9	330/29	52.3	47.3	8	7.7718	31	10.2	50.5	324.6
GP3	61.7	21.7	330/29	36.2	38.7	9	8.9356	124	4.6	53.4	348.5
Silver Bell Formation											
SV1	54.3	306.0	330/29	55.6	349.2	8	7.9281	97	5.6	80.3	184.3
SV2	82.0	306.0	330/29	63.3	43.5	6	5.7816	23	14.3	54.3	45.3
SV3	58.1	261.0	330/29	79.0	324.1	7	6.9712	209	4.2	48.4	229.9
Confidence Peak Volcaniclastic Sediments											
CS1	67.0	253.6	317/44	64.8	22.7	8		[−52,−1]	[6.4, 8.0]	67.5	292.2
CS2	64.0	203.9	305/42	72.9	51.7	12		[−57,−1]	[5.0, 6.1]	47.0	285.4
CS3	73.0	229.8	305/42	64.1	25.1	5		[−27,−2]	[10.1, 17.9]	66.5	296.2
CS4	−70.4	184.0	295/42	−29.4	197.1	15	14.4851	27	7.5	67.2	21.5
CS5	81.7	33.3	328/45	37.4	53.7	10		[−29,−1]	[7.5, 10.4]	41.2	338.1
CS6	76.2	354.1	298/39	39.1	18.1	9	8.8097	42	8.0	70.9	6.7
Tuff of Confidence Peak: Extracaldera Cooling Units											
CT1	62.4	214.5	298/65	52.3	23.1	7	6.9459	111	5.8	70.6	330.7
CT2	56.8	203.3	298/65	58.0	32.8	4	3.9699	100	9.3	62.7	315.8
CT3	72.8	223.9	315/38	69.2	45.9	8	7.9446	126	5.0	51.5	292.6
CT4	72.1	226.8	305/42	65.3	26.4	8	7.8642	52	7.8	65.0	293.9
CT5	68.6	236.3	305/42	66.8	15.3	6	5.9036	52	9.4	69.6	277.8
CT6	64.4	256.0	305/42	62.7	356.9	6		[−59,−2]	[6.2, 10.7]	78.1	237.5
Tuff of Confidence Peak: Intracaldera Fill											
CT7	73.4	4.2	325/32	45.8	36.5	8	7.9851	470	2.6	58.1	338.3
CT8	77.6	219.9	281/28	71.9	351.5	6	5.9654	144	5.6	64.9	237.5
CT9	79.9	244.7	281/28	66.6	350.1	6	5.9901	503	3.0	71.7	227.5
CT10	87.2	263.5	281/28	62.7	5.2	8	7.9522	146	4.6	77.7	266.2
Biotite Granite of El Tiro											
BG1	63.4	343.7	330/29	46.9	20.4	8	7.8919	65	6.9	71.9	346.8
BG2	58.4	302.4	330/29	59.6	353.4	6	5.9737	190	4.9	80.4	217.2
BG3	75.3	218.7	330/29	73.9	79.5	6	5.9497	99	6.8	32.8	284.2
BG4	87.0	250.8	330/29	64.0	58.7	6	5.9590	122	6.1	43.6	304.0
*Average: All Units**											
				61.2	24.0	34	31.1472	12	7.6	68.7	304.1

I and D, inclination and declination of paleomagnetic directions, uncorrected/corrected for structural tilt; strike/dip, strike and dip of unit at site; N, number of samples analyzed; R, vector sum; k, precision parameter [*Fisher*, 1953]; α_{95}, radius of 95% confidence; latitude and longitude, latitude and east longitude of virtual geomagnetic pole corrected for tilt; Average, mean corrected paleomagnetic directions. For directions resolved with demagnetization planes and Bingham statistics [*Onstott*, 1980], R is not calculated, [k_1, k_2] are the two Bingham concentration parameters for poles to the demagnetization planes, and [α_1, α_2] are the two Bingham 95% confidence limits about the mean direction.

*Not including results from the Oxide pit and from sites MP1-3 (see results).

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TABLE 2. Site-Mean Directions and Poles for Tertiary Units From the Silver Bell Mountains, Arizona

Site	Uncorrected						Pole	
	I, deg	D, deg	N	R	k	α_{95}, deg	Latitude, deg	Longitude, deg
				Andesite Flows				
TA1	−16.3	173.9	5	4.9944	709	2.9	65.2	83.2
TA2	−23.9	167.3	6	5.9907	538	2.9	67.0	101.8
TA3	−19.0	169.1	8	7.9797	345	3.0	65.2	95.0
TA4	−23.2	173.5	8	7.9857	488	2.5	68.8	86.6
TA5	−18.8	167.6	7	6.9924	785	2.2	64.5	98.0
TA6	−19.4	174.0	6	5.9971	1727	1.6	66.9	83.7
TA7	−18.9	171.9	5	4.8968	39	12.5	66.1	88.7
TA8	−24.2	170.0	8	7.9851	468	2.6	68.3	95.7
Avg 1-8	−20.5	170.9	8	7.9838	431	2.7	66.6	91.6
TA9	−42.9	163.3	5		[−220,−6]	[3.1, 10.6]	73.5	135.5
TA10	10.8	176.1	5	4.9814	215	5.2	51.9	74.9
TA11	45.2	319.3	9	8.9391	131	4.5	54.4	160.2
TA12	−53.4	189.5	3	2.9960	503	5.5	81.9	325.4
TA13	−48.2	185.3	6	5.9682	157	5.4	84.5	12.1
TA14	−59.8	192.6	5		[−146,−1]	[4.7, 5.9]	77.0	295.7
				Andesite Dike				
AD1	45.5	24.9	10	9.9113	101	4.8	67.8	346.2
				Rhyolite Dike				
RD1	48.8	30.8	7	6.8777	49	8.7	63.6	336.1
RD2	−56.4	143.5	3	2.9951	406	6.1	59.9	177.5
				Average; All Units				
	43.8	357.3	10	9.0541	10	16.5	82.8	88.4

See Table 1 footnotes.

1978; *Kirschvink*, 1980]. The statistical methods of *Fisher* [1953] were used in analyzing the within-site line directions and unit means, and Bingham statistics [*Onstott*, 1980] were used to calculate the mean direction for those sites exhibiting intersecting demagnetization planes.

Sampling and Results

The biotite granite of El Tiro was sampled at four unmineralized and relatively unaltered sites to the south and west of the El Tiro pit (Figure 2). The coarse-grained granite was sampled at three sites, and an aplite phase was sampled at site BG2. The intensities of natural remanent magnetization (NRM) for all sites averaged around 1×10^{-1} A m^{-1}, and for each site, except BG4, NRM directions were well grouped. The af demagnetization to 70 mT showed univectorial decay of magnetizations for most samples (Figure 3*a*), and thermal demagnetization to 680°C was necessary to completely demagnetize samples from site BG4. Secondary, perhaps lightning-induced, magnetizations were generally removed from samples of site BG4, and three samples from this site that could not be magnetically "cleaned" were omitted from the site-mean calculation. Site-mean directions (Table 1) are shown in Figure 4 both before and after structural correction.

Samples from the tuff of Confidence Peak were collected at six sites in extracaldera cooling units within the western Silver Bell Mountains and at four sites in intracaldera fill within the main Silver Bell range (Figure 2). Site-averaged NRM intensities for the extracaldera units ranged from 1 A m^{-1} to 8×10^{-2} A m^{-1} and the dispersion of NRM directions ranged from well grouped to highly scattered. The high NRM intensities and dispersed directions are inferred to be the result of lightning-induced magnetizations [*Graham*, 1961; *Cox*, 1961]. The af demagnetization to 100 mT was generally successful in removing these secondary magnetizations and in isolating a characteristic magnetization for each site (Figure 3*b*). One sample at each of sites CT1, CT4, and CT5 and four samples at site CT2 were completely overprinted by spurious magnetizations and were omitted from the site averages. Demagnetization planes were required to determine a site-mean direction for site CT6, and thermal demagnetization, to 680°C, was performed on samples from sites CT3 and CT6. The demagnetization behavior for samples from the four intracaldera sites was less complicated. NRM intensities ranged from 8 A m^{-1} to 1×10^{-1} A m^{-1}, but the NRM directions were well grouped. The af demagnetization to 90 mT showed single-component magnetizations.

Volcaniclastic sedimentary rocks interbedded with outflow pyroclastic and air fall deposits of the tuff of Confidence Peak were sampled in the western Silver Bell Mountains at seven sites. Intensities of NRM ranged as high as 20 A m^{-1} but were generally between 1×10^{-2} A m^{-1} and 3×10^{-1} A m^{-1}. Directions of NRM tended to be scattered but were better grouped for sites CS4, CS5, and CS6. Thermal demagnetization to temperatures between 600°C and 670°C revealed multicomponent magnetizations (Figure 3*c*), and demagnetization planes were used at most sites to calculate site-mean directions. The sampling sites usually included several meters of section, and at sites CS4, CS5, and CS6, samples with characteristic magnetizations of both polarities were observed. Site CS7 exhibited the highest NRM intensities and dispersed NRM directions, indicative of lightning-induced magnetizations. Demagnetization treatments failed

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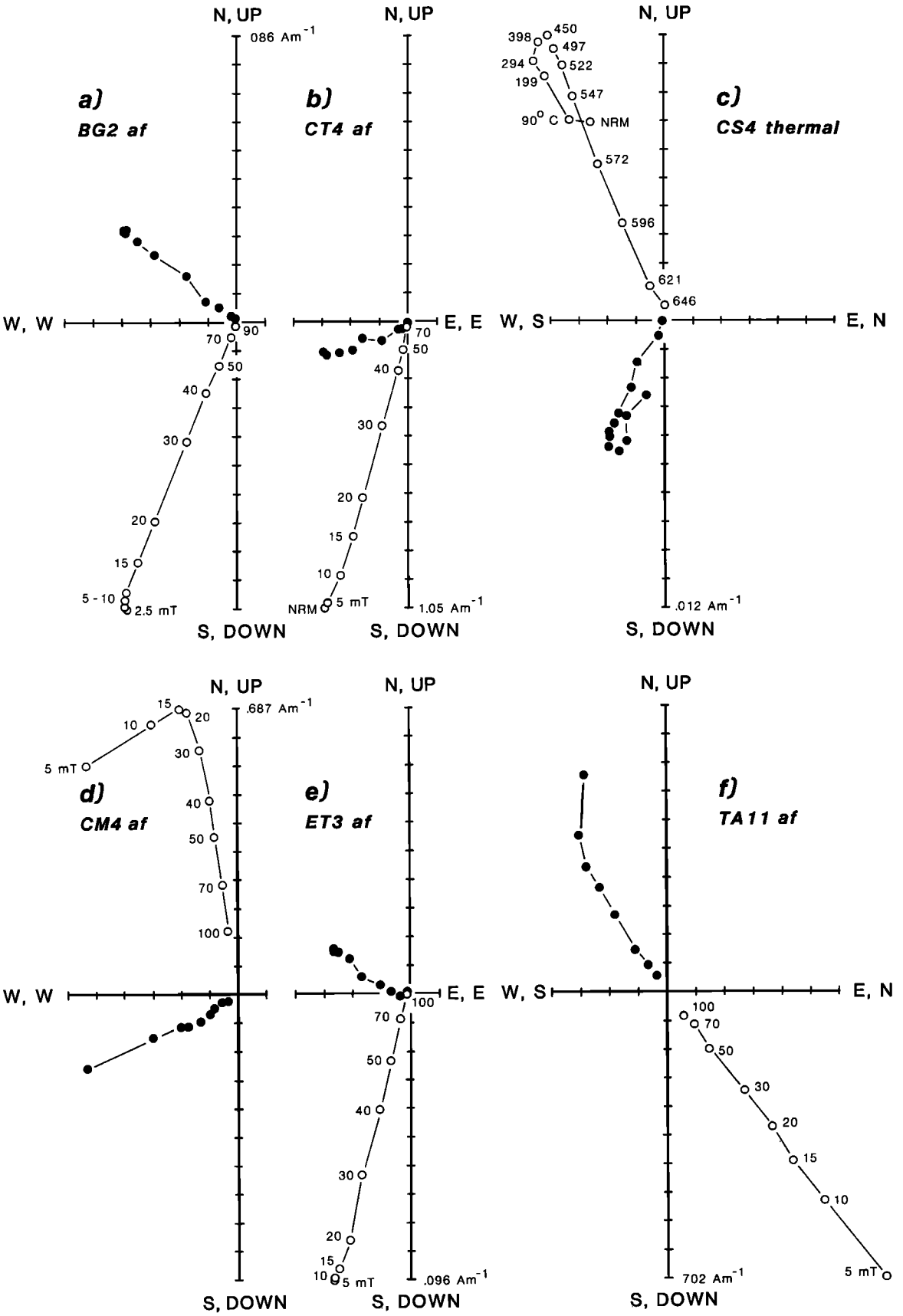


Fig. 3. Orthogonal projections of uncorrected demagnetization vector endpoints for samples from (*a*) biotite granite of El Tiro, (*b*) tuff of Confidence Peak, (*c*) Confidence Peak volcaniclastic sedimentary rocks, (*d*) Cat Mountain Rhyolite, (*e*) quartz monzodiorite porphyry in the El Tiro pit, and (*f*) Tertiary andesite. The af denotes alternating field demagnetizations, and thermal denotes thermal demagnetizations. Solid symbols indicate projections of vector endpoints onto the horizontal plane and open symbols onto the vertical plane.

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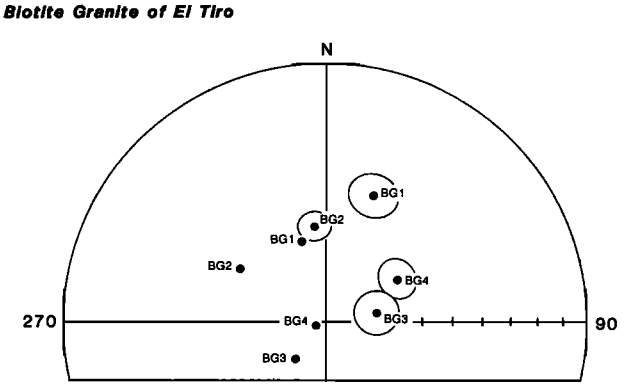


Fig. 4. Mean directions of sites corrected for structural tilt (with 95% confidence limits) and uncorrected for tilt (without confidence limits) for the biotite granite of El Tiro. Diagram is an equal-area plot where solid symbols indicate projection from the lower hemisphere.

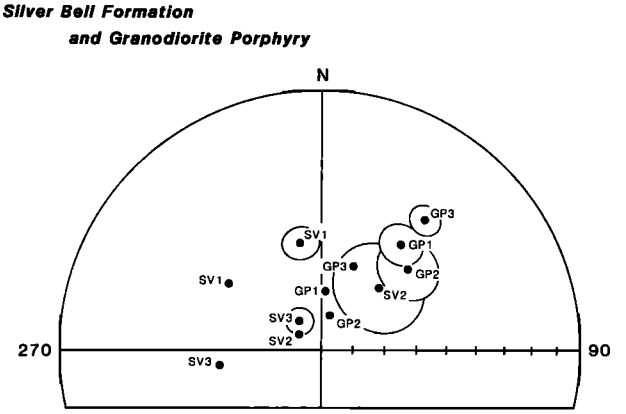


Fig. 6. Mean directions of sites corrected for structural tilt (with 95% confidence limits) and uncorrected for tilt (without confidence limits) for the volcanics of the Silver Bell Formation and granodiorite porphyry pluton. Diagram is an equal-area plot where solid symbols indicate projection from the lower hemisphere.

to isolate a consistent characteristic magnetization for this site, and results from it are not included in Table 1. Three samples with inconsistent magnetizations were also omitted from the mean for site CS3. Site-mean directions for the tuff of Confidence Peak and interbedded sedimentary units (Table 1) are shown in Figure 5. Before structural correction the steep southwesterly directions are unlike any for North America, implying that they do not represent a more recent (Tertiary) remagnetization. The corrected directions have inclinations consistent with those for stable North America, but the declinations are more easterly suggesting a clockwise rotation of the Silver Bell Mountains.

Two sites in porphyritic andesite flows and one site in a quartz monzodiorite porphyry dike (SV3) were located within the Silver Bell Formation east of the Silver Bell Mountains (Figure 2). Three sites were also situated in a granodiorite porphyry pluton, most likely related to the Silver Bell Formation, exposed in a stream wash west of Ragged Top. NRM intensities for these units were similar and averaged around 4×10^{-1} A m^{-1}. NRM directions were generally well grouped, and demagnetization of the samples was straightforward. Heating to between 650°C and 680°C of samples from the Silver Bell Formation and af demagnetization to 90 mT of samples from the granodiorite porphyry pluton yielded characteristic magnetizations with directions similar to the NRM directions. Mean directions for these sites (Table 1) are shown in Figure 6, and the corrected directions are consistent with those for the other Upper Cretaceous units in the region.

The Cat Mountain Rhyolite was sampled at five sites in the main and western Silver Bell Mountains, and the underlying rhyolitic lava flows of the Mount Lord Volcanics were

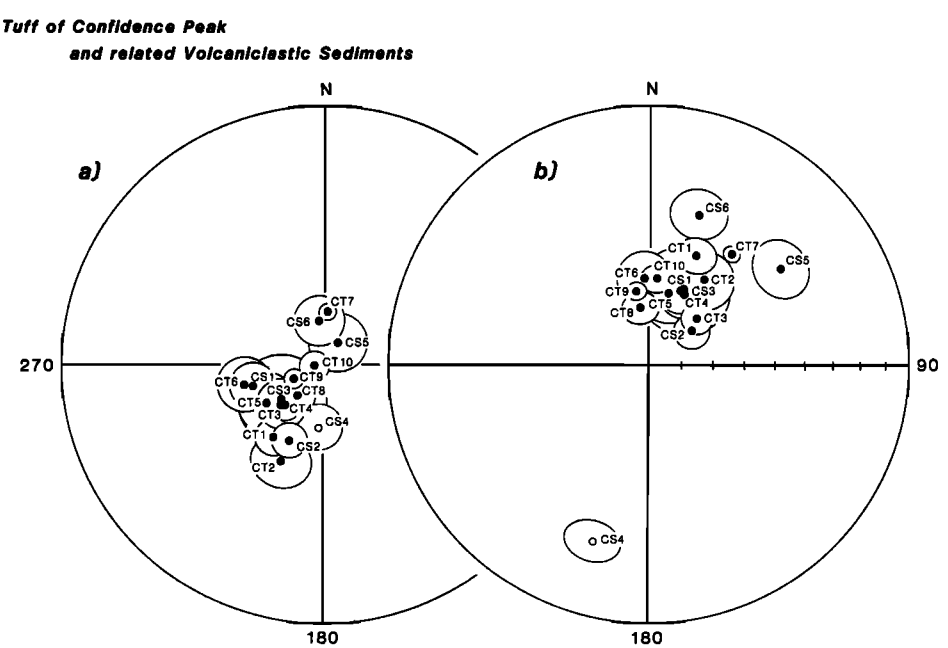


Fig. 5. Mean directions of sites (*a*) uncorrected and (*b*) corrected (with 95% confidence limits) for structural tilt for the tuff of Confidence Peak and volcaniclastic sedimentary rocks. Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere and open symbols from the upper hemisphere.

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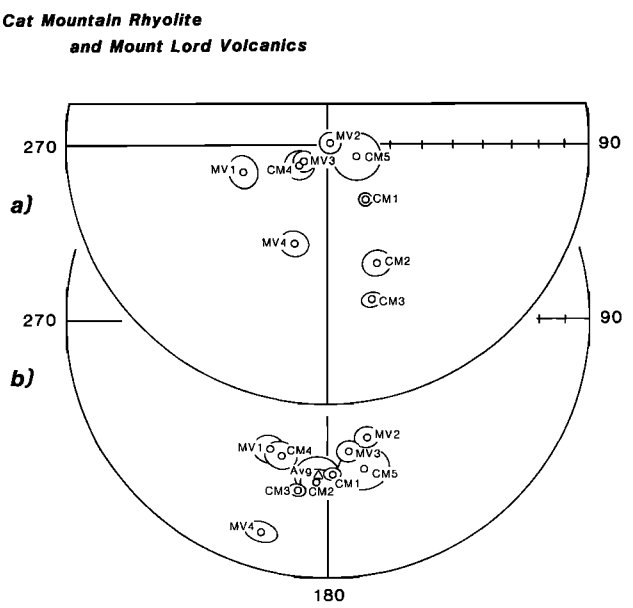


Fig. 7. Mean directions of sites (*a*) uncorrected and (*b*) corrected for structural tilt (with 95% confidence limits) for the Cat Mountain Rhyolite and Mount Lord Volcanic rocks. Avg (triangle) is the average direction for the Cat Mountain Rhyolite. Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere and open symbols from the upper hemisphere.

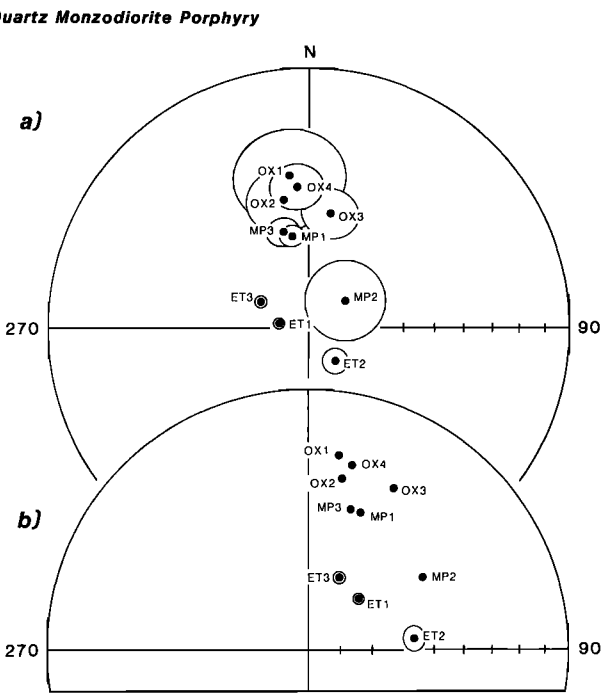


Fig. 8. Mean directions of sites (*a*) uncorrected and (*b*) corrected for structural tilt (with 95% confidence limits) for the quartz monzodiorite porphyry suite. Directions in Figure 8*b* without confidence limits were omitted from the overall Late Cretaceous average. Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere.

sampled at four sites (Figure 2). The NRM directions for the Cat Mountain Rhyolite and Mount Lord Volcanics tend to be highly scattered (except for sites MV3, MV4, and CM4) and the NRM intensities range from 2×10^{-1} A m^{-1} to 10 A m^{-1}, both being indicative of lightning-induced magnetizations. The af demagnetization to between 70 and 100 mT was generally successful in removing the lightning-induced overprints and in isolating characteristic magnetizations in the rock samples (Figure 3*d*). Two samples from sites CM1, CM2, and CM3 and one sample from site MV2 were not included in the site-mean calculations because they had been completely remagnetized by lightning strikes. Demagnetization planes were used to determine the site-mean direction for site CM5. Site-mean directions for the Cat Mountain Rhyolite (Figure 7 and Table 1) are significantly better grouped at the 95% confidence level (k_{unc} = 13, k_{cor} = 64 [*Watson and Irving*, 1957]) after correction for the attitudes of its eutaxitic structure, implying that the rhyolitic tuff was magnetized prior to structural deformation and that it carries a primary thermoremanent magnetization. The greater dispersion in declination than in inclination of corrected directions for the Cat Mountain Rhyolite is probably the result of small between-site vertical axis rotations. These rotations are inconsistent with a hypothesis of oroclinal bending in the Silver Bell Mountains and are most likely caused by local movements within the regional deformation.

The quartz monzodiorite porphyry suite was sampled at 10 sites. Four sites are located within the Oxide pit, three sites are located within the El Tiro pit, and three sites are situated nearby (Figure 2). NRM intensities for all sites are around 2×10^{-1} A m^{-1}, and the NRM directions tend to be well grouped. The af demagnetization of samples from the Oxide pit removed a single component of magnetization by 20–30 mT. Continued demagnetization to 70 mT provides evidence of a second higher-coercivity component of magnetization which cannot be isolated because it is almost completely overprinted. The lower-coercivity component was removed by thermal demagnetization to 584°C in samples from sites OX3 and OX4 but was unstable to thermal demagnetization in samples from sites OX1 and OX2. One sample from site OX1 and three samples from site OX4 were discarded because of probable lightning-induced overprints. The similarity between the direction of magnetization determined for rocks of the Oxide pit and the Tertiary reference direction implies that the Oxide pit magnetization is a thermoviscous overprint related to local Tertiary igneous activity. These site means are not included in the overall Late Cretaceous average (Table 1). On the other hand, samples from the El Tiro pit show univectorial components of magnetization, after af treatment to 100 mT (Figure 3*e*) or heating to 584°C, that are most likely a primary thermoremanence. These magnetization directions are unlike any reference directions for North America at Silver Bell and are consistent with those from other Upper Cretaceous units in the region (Figure 8 and Table 1). Samples from the three intervening sites in the quartz monzodiorite porphyry suite also yield directions (MP1, MP3) and exhibit demagnetization behavior (MP2) indicative of secondary overprints; thus these results have been omitted from the overall Late Cretaceous average.

Tertiary andesite and basaltic andesite flows were sampled at fifteen sites east of the Silver Bell Mountains (Figure 2). Dikes of Tertiary age were also sampled at three sites in the main Silver Bell range; an andesite dike was sampled in the El Tiro pit (AD1), and rhyolite dikes were sampled north (RD2) and northwest (RD1) of the Oxide pit. As with samples of other rock units in the area, the NRM of about

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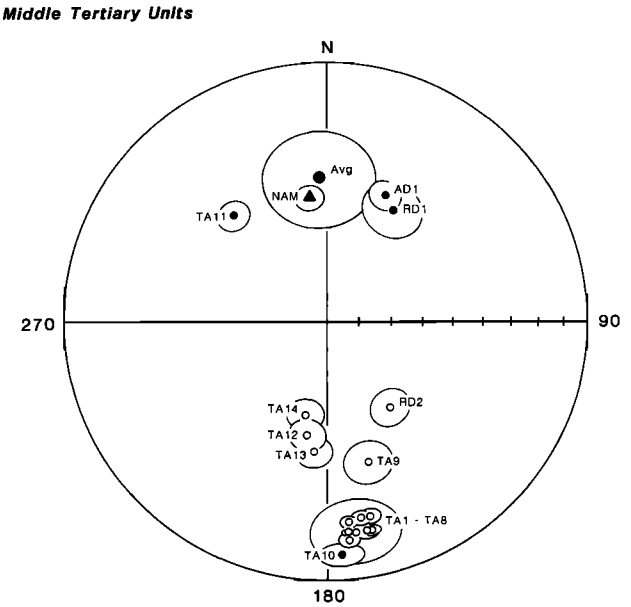


Fig. 9. Mean directions of sites (with 95% confidence limits) for the Tertiary units. Larger symbols labeled Avg (circle) and NAM (triangle) indicate average directions for the Silver Bell units and the direction at Silver Bell calculated from data for units of the North American craton. Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere and open symbols from the upper hemisphere.

half the Tertiary samples is influenced by lightning-induced magnetizations. NRM directions for these samples are scattered, and the NRM intensities tend to be an order of magnitude or two greater than the average for the unaffected sites (4×10^{-1} A m^{-1}). The af demagnetization to 100 mT was generally successful in removing the overprint magnetizations and in isolating characteristic magnetizations (Figure 3*f*). However, the characteristic magnetizations had been completely obliterated by lightning in all samples from site TA15, in five samples from sites RD2 and TA12, in three samples from site TA14, in two samples from sites TA1, TA9, and TA10, and in one sample from sites TA7 and RD1. Demagnetization planes were used to determine mean directions for sites TA9 and TA14. Sites with well-grouped NRM directions showed univectorial characteristic magnetizations upon demagnetization. The site-mean directions for the Tertiary units (Table 2) are shown in Figure 9. The tight cluster of mean directions for sites TA1 through TA8 implies that a limited number of flows, erupted over a short period of time relative to geomagnetic secular variation, were sampled at these sites. Results for these eight sites have been averaged because they probably represent a record of the geomagnetic field over a short interval of time, comparable to that represented by a single flow.

Discussion and Conclusions

Site-mean directions for the Upper Cretaceous units of the Silver Bell Mountains are shown in Figure 10 both before and after correction for structural tilting. Due to the similarity of structural corrections for all rock units, the overall dispersion of the corrected site-mean directions is not significantly changed (k_{unc} = 15, k_{cor} = 12). The angular standard deviation of the 34 corrected virtual geomagnetic poles (S = 20.6°) is larger than the value expected for the latitude of Silver Bell (S = 15.5° ± 1°) based on models of recent paleosecular variation [e.g., *McFadden et al.*, 1988]. Errors in the experimental procedure, undetected remagnetizations, within-site dispersion, excursions of the geomagnetic field, and incomplete structural corrections may also contribute to the observed dispersion. Of these, incomplete (e.g., Cat Mountain Rhyolite, see results section) or potentially inaccurate (e.g., intrusive units) structural corrections are probably responsible for most of the excessive dispersion. These errors are apparently nonsystematic, so an

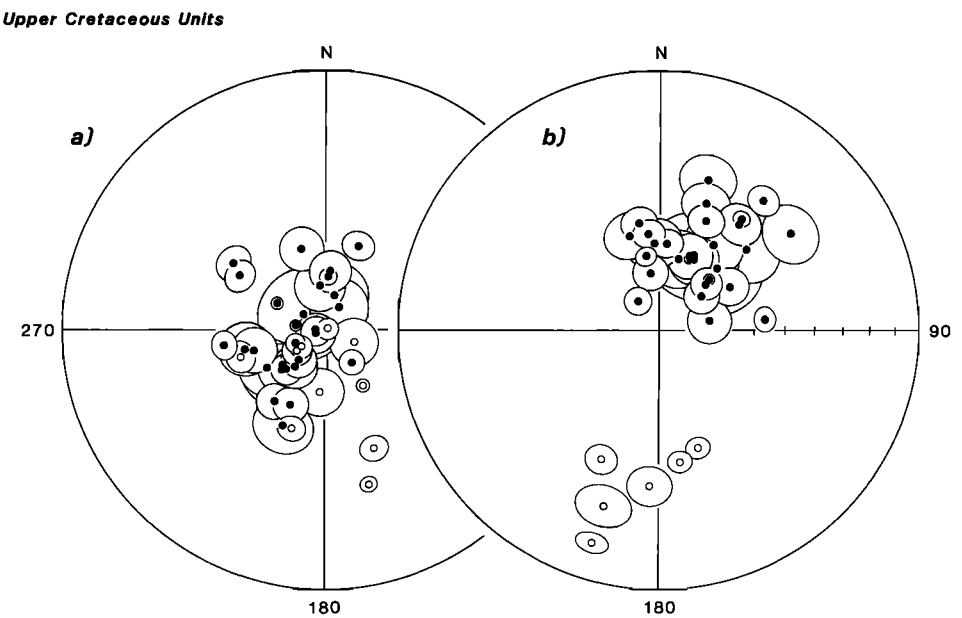


Fig. 10. Mean directions of sites (*a*) uncorrected and (*b*) corrected for structural tilt (with 95% confidence limits) for the Upper Cretaceous units at Silver Bell. Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere and open symbols from the upper hemisphere.

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TABLE 3. Rotations Between Reference Directions for North America and Mean Directions for the Silver Bell Mountains, Arizona

Rock Units	I, deg	D, deg	N	R	k	α_{95}, deg	d, deg	r, deg	Reference
				Tertiary					
North America	50.2	352.2	102	94.6429	14	3.9			1
All units	43.8	357.3	10	9.0541	10	16.5	6.4 ± 13.3	5.1 ± 18.5	2
				Upper Cretaceous					
North America	58.8	342.3	7	6.9036	83	7.7			3
All units	61.2	24.0	34	31.1472	12	7.6	−2.4 ± 7.6	41.7 ± 14.3	2

See Table 1 footnotes; d, r, poleward displacement and clockwise rotation of site with respect to cratonic North America, with 95% confidence limits after *Beck* [1980], as modified by *Demarest* [1983]. References: 1, direction at Silver Bell calculated from *Diehl et al.*'s [1983] Table 6, Oligocene-early Miocene entry; 2, this study; 3, direction at Silver Bell calculated from *Irving and Irving*'s [1982] Table 1, entries 33–39.

overall mean direction is most likely not significantly affected.

The overall mean direction for the Upper Cretaceous units of the Silver Bell Mountains (I = 61.2°, D = 24.0°, α_{95} = 7.6°) is distinctly different from averaged directions for contemporaneous units from elsewhere on the North American continent (I = 58.8°, D = 342.3°, α_{95} = 7.7°; Table 3 and Figure 11). This difference indicates a Late Cretaceous paleolatitude anomaly of −2.4° ± 7.6° (not significant at the 95% confidence level) and a declination anomaly of 41.7° ± 14.3° (Table 3). Because the geomagnetic field direction is relatively well known for Late Cretaceous time in North America, we infer that the declination anomaly is most likely caused by clockwise rotation in the Silver Bell Mountains with respect to North America since Late Cretaceous time. Two explanations are possible for the apparent clockwise rotation in the Silver Bell Mountains: (1) association with movement before 26 Ma along WNW trending strike-slip faults bounding the Silver Bell Mountains to the north and south, or (2) association with post-26 Ma extension related to the Rincon–Santa Catalina–Tortolita core complexes situated nearby to the east. Although preliminary, the average direction of the Oligocene to early Miocene paleomagnetic data from the Silver Bell region (I = 43.8°, D = 357.3°, α_{95} = 16.5°) is not significantly different from a contemporaneous

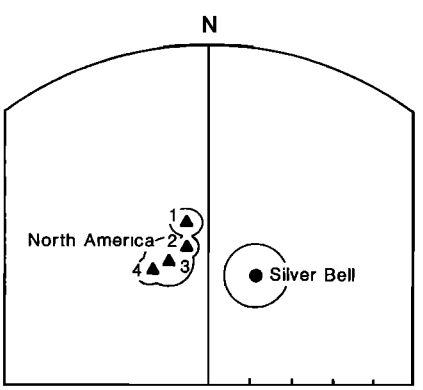


Fig. 11. Mean and reference directions (with 95% confidence limits) for Upper Cretaceous units at Silver Bell (circle) and for the North American continent (triangles). 1, Oligocene to early Miocene direction calculated from *Diehl et al.* [1983] (see Table 3); 2, Paleocene direction calculated from *Diehl et al.* [1983]; 3, Late Cretaceous direction calculated from *Irving and Irving* [1982] (see Table 3); and 4, Cretaceous direction calculated from *Irving and Irving* [1982]. Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere.

direction for North America (I = 50.2°, D = 352.2°, α_{95} = 3.9°; Figure 9 and Table 3). Moreover, paleomagnetic data from Tertiary units near Ajo, Arizona (*Hagstrum et al.*, 1987*b*], and across western Arizona [*Calderone*, 1988] indicate no detectable tectonic translations or rotations in the region. These data imply that the 40° clockwise rotation of the Silver Bell Mountains occurred before 26 Ma and was probably associated with strike-slip movement along the Ragged Top and related WNW trending faults bounding the Silver Bell Mountain block.

Previously determined paleomagnetic data for other Mesozoic rocks in southern Arizona suggest that the significant clockwise rotation of the Silver Bell Mountains is a local phenomenon. Paleomagnetic directions [*Vugteveen et al.*, 1981] for the Upper Cretaceous Roskruge Volcanics (≃ 72 Ma) from the nearby Roskruge Mountains (Figure 1) show no declination anomaly with respect to a contemporaneous North American direction (Table 3); however, the average inclination is too shallow by about 20°. This inclination anomaly may reflect rapid apparent polar wander in Late Cretaceous to early Tertiary time but could also result from incomplete averaging of secular variation (SV). Although remanent magnetizations of both polarities are observed in samples of the Roskruge Volcanics, the small number of sites (N = 9) and small angular standard deviation of virtual geomagnetic poles (S = 9.0°) suggest that SV has not been averaged. Similarly, paleomagnetic data for the lower Paleocene Gringo Gulch Volcanics [*Vugteveen et al.*, 1981] are consistent with a Paleocene pole for North America (*Diehl et al.*, 1983] but apparently do not average SV. The 25 virtual geomagnetic poles are of uniform reversed polarity and tightly grouped (S = 4.2°), implying that the sampled lava flows were erupted over a short interval of time relative to SV. Thus an inference that southern Arizona has remained entirely stationary with respect to stable North America, based on the apparent concordance of these data, is questionable.

Paleomagnetic data from studies of Jurassic volcanic rocks also bear on the tectonic stability of southern Arizona. *Kluth et al.* [1982] analyzed samples from 15 units (S = 13.2°) in Upper Jurassic volcanic flows of the Upper Jurassic to Lower Cretaceous Glance Conglomerate, exposed within the Canelo Hills (Figure 1), to determine a reference Jurassic paleopole for stable North America. The consistency of this paleopole with the Jurassic APW path for North America implies that little, if any, relative translation or rotation of the region has occurred since Jurassic time. *May et al.* [1986] report a paleopole, calculated from 12 sites (S = 11.5°) in

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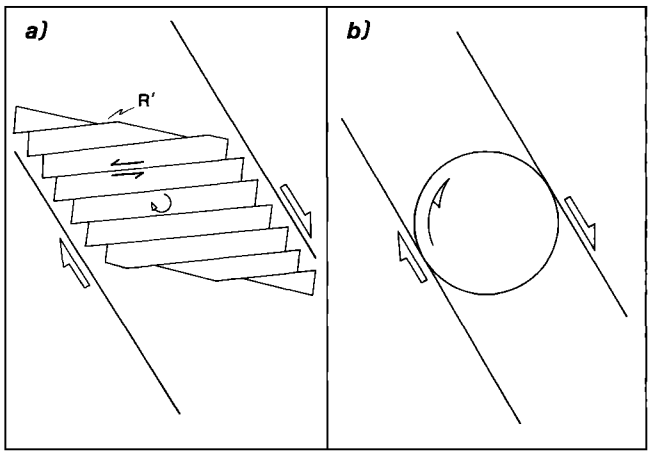


Fig. 12. Diagrams of tectonic models for clockwise rotation [after *Wells and Heller*, 1988]. (*a*) rigid-blocks bounded by strike-slip faults (R', Riedel shear faults), and *b*) "ball-bearing" model for the Silver Bell Mountains.

Jurassic volcanic rocks from the Patagonia Mountains (Figure 1), that is also consistent with the APW path for North America. However, two sites were omitted from this data set because their more westerly declinations were thought to reflect local counterclockwise rotation ($\simeq$ 40°) associated with strike-slip motion on a nearby fault. Another data set of 28 site means (S = 15.1°) for a sequence of volcanic rocks stratigraphically beneath the Glance Conglomerate in the Canelo Hills [*May and Butler*, 1987] is significantly anomalous in declination, suggesting clockwise rotation of these rocks $\simeq$ 35° relative to site locations in the overlying Glance Conglomerate [*Kluth et al.*, 1982]. *May and Butler* [1987] interpret this "anomalous" data set from the Canelo Hills as a record of nondipole behavior of the Jurassic geomagnetic field, but note that the easterly declination could also result from clockwise rotation of the area in Late Jurassic time prior to deposition of the overlying Glance Conglomerate. The level of dispersion for each of these Jurassic data sets is reasonably consistent with a model value of SV (S = 15.5° [*McFadden et al.*, 1988]), suggesting that SV has been averaged and that the preservation of short-duration nondipole behavior of the geomagnetic field is unlikely. Because sites of the "anomalous" data set are located for the most part in separate fault blocks south of sites in the Glance Conglomerate, another possible explanation for the difference in declination is younger clockwise rotation of the southernmost Canelo Hills relative to the northern sites in the Glance Conglomerate.

Discordant paleomagnetic declinations within regional shear zones have been explained by tectonic models in which domains of rigid crustal blocks, bounded by parallel strike-slip faults (Figure 12*a*), rotate along with the faults [*Freund*, 1974; *Ron et al.*, 1984]. This model has been used to explain rotations in southern California [*Luyendyk et al.*, 1980] and southwestern Washington [*Wells and Coe*, 1985], where the external shear couple is inferred to have been between the North American plate and an oceanic plate to the west. This model is apparently not applicable to the southern Basin and Range province in Arizona as a whole, because it predicts that adjacent blocks within the shear zone, assuming the strike-slip faults remain parallel, would undergo similar amounts of rotation. In Arizona, rotated blocks (Silver Bell, southern Canelo Hills?) are adjacent to those that appear unrotated (Roskruge Mountains, northern Canelo Hills). Another model that better explains the rotated paleomagnetic data from the Silver Bell Mountains is one in which the mountain range rotated as a rigid block (Figure 12*b*) between the parallel transcurrent faults bounding the range to the north and south [e.g., *Fitch*, 1972; *Beck*, 1980].

At present, the available paleomagnetic data for southern Arizona are insufficient to constrain a tectonic model for deformation of the southern Basin and Range province in Late Cretaceous and early Tertiary time that accounts for the observed vertical axis rotation in the Silver Bell Mountains. However, the paleomagnetic data for the Colorado Plateau [*Steiner*, 1986; *Bryan and Gordon*, 1986] and the Baja California Peninsula [e.g., *Hagstrum et al.*, 1985, 1987*a*] imply that southwestern North America was subjected to NW trending right-lateral shear in Late Cretaceous to early Tertiary time [e.g., *Beck*, 1986] consistent with the clockwise rotation in the Silver Bell Mountains. Thus, it is tentatively suggested that WNW trending right-lateral strike-slip faults, associated with clockwise vertical axis rotations, played a significant role in the deformation of southern Arizona at this time. In contrast, the regional orientation of least principal stress direction inferred from fracture and dike patterns [*Rehrig and Heidrick*, 1976; *Heidrick and Titley*, 1982] implies a NW trending left-lateral shear couple for the region in Late Cretaceous and early Tertiary time. A portion of these data are for the Silver Bell district which cannot be representative of the paleostress direction at this time because of the subsequent rotation documented in this study. More paleomagnetic data are needed for southern Arizona to determine the regional tectonic implications of local vertical axis rotations.

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